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Generative artificial intelligence in pedagogical practices: a systematic review of empirical studies (2022–2024)

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ABSTRACT

This systematic review explores how generative artificial intelligence (GenAI) technologies have been employed within pedagogical practices from 2022 to 2024, uncovering specific applications, benefits and challenges. This review draws on 28 empirical studies from top high-impact factor journals indexed in the Web of Science under 'Education & Educational Research' and 'Education, Special' (Journal Citation Reports). The analysis reveals that GenAI could improve instructional efficiency by providing prompt feedback and personalized learning materials, enhancing learners' engagement and motivation, catalyzing idea generation and developing essential generic skills such as critical thinking and problem-solving. However, key challenges arise concerning potential inaccuracies, bias, threats to academic integrity and questions regarding GenAI's overall efficacy in teaching and learning. This study suggests that teachers should wisely employ GenAI as a supplementary device, optimizing its advantages and addressing its drawbacks. This shows how important it is for teachers to keep learning and how important it is to look into the long-term effects, untapped uses and compatibility of GenAI with the Sustainable Development Goals, especially when trying to improve education.

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Introduction

As interest in generative artificial intelligence (GenAI) surges, particularly following the release of ChatGPT in the latter half of 2022, the field of pedagogical practices has begun to experience notable shifts (Bahroun et al., 2023). This paradigm shift may disrupt traditional instructional paradigms while ushering in novel opportunities and challenges for educators, students and policymakers (Lai et al., 2023). Renowned for its capacity to emulate human creativity and intellect, GenAI, facilitated by Gemini, exhibits remarkable capabilities in generating textual, image and audio content, thereby enabling personalized instructional strategies, interactive learning environments and innovative content creation (Imran & Almusharraf, 2024). Multiple studies indicate that these features can reduce teachers' workload, respond to diverse learner needs and improve engagement in instructional activities (Binhammad et al., 2024; Essien et al., 2024; Hashem et al., 2023).

Although the inception of GenAI can be traced back to the utilization of chatbots in the 1960s, ELIZA, a computer software, was created to replicate the conversational skills of a psychiatrist (Pham et al., 2022). The advent of generative adversarial networks (GANs) in 2014 marked a significant milestone. By leveraging GANs, researchers have achieved breakthroughs in generating realistic images, films and audio content (Abdel-Basset et al., 2023). The widespread accessibility of ChatGPT by 2022, developed by OpenAI, has democratized access to this technology, showcasing its prowess in comprehending

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intricate queries and furnishing timely, dependable and pertinent responses. Recent advancements in GenAI by Ebert and Louridas (2023), P. Zhang and Kamel Boulos (2023) and Chuma and De Oliveira (2023) underscore the transformative potential of this technology across diverse instructional and professional spheres, spanning from higher education to lifelong learning initiatives.

Documented potential in enhancing teaching and learning methodologies, integrating GenAI in instructional practices assumes paramount significance. As Kasneci et al. (2023) posited, this technology streamlines the creation of personalized educational content, augments instructional strategies and fosters interactive engagement with learning materials, thereby conferring tangible benefits to the instructional domain. By leveraging these capabilities, educators can deliver more tailored instruction and reduce manual workload, ultimately benefiting students and teachers. However, amidst these promises, GenAI encounters multifaceted challenges. Concerns about academic integrity, AI accuracy, overreliance, digital divide, privacy and security loom large. Both Niloy et al. (2024) and Perkins and Roe (2024) caution against the unchecked integration of GenAI into pedagogical practices, suggesting that its adoption may potentially exacerbate existing instructional challenges rather than resolve them.

While the intersection of artificial intelligence and education has garnered substantial scholarly attention in recent years (Duart, 2024; Filgueiras, 2024; Kong et al., 2025; Rahiman & Kodikal, 2024), the empirical exploration of GenAI technologies such as Deepseek, Gemini, Claude, LLaMA and other large language models (LLMs), remains relatively nascent. Existing studies separately examine GenAI's capacity to personalize instruction (Pesovski et al., 2024), enhance motivation and engagement in learning (S. Chan et al., 2025; Minh et al., 2024), improve learning outcomes (Xiaoyu et al., 2024) and foster critical thinking and problem-solving skills (Zhou et al., 2024). For instance, Pesovski et al. (2024) demonstrated that GenAI personalizes instruction by tailoring course materials to individual preferences, increasing engagement through relatable formats. Minh et al. (2024) highlighted LLMs' ability to enhance students' self-learning motivation by employing GenAI tools such as Gemini, Bing Chat and Claude AI.

Similarly, S. Chan et al. (2025) found that GenAI feedback significantly enhanced engagement and motivation during writing revisions. Xiaoyu et al. (2024) observed notable improvements in learning outcomes when AI tools were paired with strategies for critical evaluation. Finally, Zhou et al. (2024) identified self-regulation as a key mediator between AI ease of use and developing critical thinking and problem-solving skills. These findings underscore GenAI's transformative potential across various instructional domains. However, the broader scope of GenAI research in pedagogy remains underexplored, particularly regarding its growing ability to mimic human creativity, which often falls outside the focus of existing analyses and highlights the need for more targeted investigations.

Against this backdrop, this study fills a significant lacuna in the literature by narrowly focusing on the distinctive ability of GenAI to emulate human creativity, a facet frequently overlooked in prior AI studies. Unlike previous systematic reviews that treated GenAI as one of many AI-driven educational technologies, this study focuses exclusively on its pedagogical role in the post-ChatGPT era. By analyzing empirical studies from 2022 to 2024, this review captures the rapid evolution of GenAI tools and their immediate educational impact, an area not comprehensively explored in earlier work. This refined focus enables a deeper investigation into GenAI's distinct affordances, particularly its role in creative content generation, AI-assisted instruction and personalized learning. While prior research has largely examined AI-driven automation and adaptive learning (Dutta et al., 2024; Khine, 2024; Strielkowski et al., 2024), this review extends the discussion by exploring how GenAI improves instructional efficiency, enhances engagement and motivation, inspires main idea and develops generic skill. These insights contribute to the ongoing discourse on AI as a collaborative partner in education, highlighting both its potential to enhance creativity and the challenges it poses for academic integrity and instructional design.

By delving into empirical applications within the specified timeframe (2022–2024) and post-major technological breakthroughs like the release of ChatGPT, this study provides fresh insights into the evolving landscape of GenAI in pedagogical practices while identifying uncharted research avenues. Moreover, this study addresses ethical dilemmas, and academic integrity concerns specific to GenAI, underscoring the importance of originality and reliability in AI-generated content. Furthermore, by establishing a nexus with the Sustainable Development Goals, this study offers a structured framework for evaluating the contributions and challenges of GenAI toward achieving global instructional objectives.

By employing a rigorous and systematic approach to literature analysis, this study critically examines 28 studies to evaluate the efficacy and implications of GenAI in pedagogical practices. Beyond contributing to ongoing scholarly discourse, this comprehensive review furnishes invaluable insights for informing future policy formulations, research agendas and instructional strategies within educational technology.

Research purpose and research questions

Rapid GenAI advancements have triggered significant interest in its instructional applications. Although prior research has broadly examined AI in educational settings, the unique capacities of GenAI, such as text generation, image creation and automated feedback (Bansal et al., 2024; Huang, 2024; S. S. Lee & Moore, 2024), remain insufficiently explored from an empirical pedagogical standpoint. Consequently, this study seeks to (i) systematically analyze existing empirical research on GenAI in instructional contexts, (ii) identify its potential applications and benefits in learning and teaching practices and (iii) uncover research gaps for future investigation in pedagogy.

Building upon these aims, the subsequent research questions will guide our review:

1. What distinctive features define the current empirical landscape of GenAI in pedagogical practices, including research countries, publication sources, research objectives, methodological approaches, educational levels, application domains and specific GenAI platforms and applications?
2. What is the potential landscape for implementing GenAI in pedagogical practices?
3. What untapped areas warrant future investigation into the role of GenAI in learning and instruction?

Methods

Research design

This study comprehensively examines the current empirical literature concerning GenAI's role in pedagogical practices. The empirical data in this investigation were sourced from 28 peer-reviewed journal articles published between 2022 and 2024, reflecting the rapid proliferation of GenAI since November 2022 (Bahroun et al., 2023). Acknowledging the importance of transparent reporting and methodological rigor in knowledge synthesis, the study adheres to the PRISMA guidelines advocated by Sarkis-Onofre et al. (2021). These guidelines were designed to aid authors in accurately documenting various knowledge synthesis methodologies, including scoping, systematic and review processes, thereby enhancing research transparency and minimizing bias.

Furthermore, the study employs rigorous content and thematic analysis techniques to ensure the comprehensive and systematic analysis of empirical research on GenAI in pedagogical practices (Braun et al., 2019; Krippendorff, 2019). This analytical approach involves transforming textual content into relevant codes, topics and categories, facilitating the synthesis of disparate pieces of information into coherent and meaningful themes. By adopting such methodological rigor, the study aims to enhance the credibility and trustworthiness of its findings, thereby contributing to the robustness of the research in this burgeoning field.

A structured multi-phase approach was employed to systematically categorize and analyze the reviewed studies. Initially, a predefined coding framework was developed based on key study characteristics, including research objectives, methodological approaches, participant demographics, GenAI platforms and educational contexts. Informed by prior literature and refined iteratively, this framework ensured consistency and alignment with the study's research questions. Studies were first categorized into broad thematic areas, pedagogical applications, instructional effectiveness and ethical considerations, before being further segmented into specific subthemes to capture nuanced trends and challenges.

To enhance the accuracy and reliability of the thematic coding process, two independent researchers conducted coding separately, followed by a cross-validation procedure to reconcile discrepancies. Any disagreements were resolved through discussion, and if necessary, a third researcher provided

adjudication. An audit trail was maintained to document coding decisions, ensuring transparency in thematic evolution.

To minimize bias, strict inclusion and exclusion criteria were applied, ensuring that only empirical research directly related to GenAI in education was analyzed. Additionally, triangulation techniques were used to integrate findings from diverse sources, reducing the risk of interpretive distortions. Reflexivity measures, such as maintaining analytic memos, helped track emerging insights, potential biases and alternative interpretations. Finally, adherence to PRISMA guidelines facilitated transparent methodological reporting, reinforcing the study's credibility and reproducibility.

Search process

In accordance with Moher et al. (2009), PRISMA represents a structured approach tailored for systematic reviews and meta-analyses, thereby bolstering the reliability and consistency of research outcomes. As such, this study adhered to the PRISMA guidelines during its literature screening process, ensuring methodological rigor and transparency. Three systematic review steps were meticulously implemented to identify pertinent literature. Initially, the researchers validated the selected keywords to guide the search process. Subsequently, publications underwent rigorous scrutiny based on pre-established criteria to determine inclusion or exclusion. The final phase encompassed the assessment of eligibility and inclusion of empirical studies, culminating in a comprehensive selection process that upheld the principles of systematic review methodology. Through the diligent application of these systematic review steps, this study endeavored to enhance the validity and comprehensiveness of its literature review, thereby facilitating robust analysis and interpretation of findings.

Identification of sources

In preparation for the literature search, relevant terms were extracted from previous studies by Bahroun et al. (2023) and Bozkurt (2023). Employing Boolean operators, such as 'AND', the following terms were combined: 'generative AI*', 'generative artificial intelligence*', 'GenAI*' and 'Large language model*', with 'instruction*', 'teaching and learning*' and 'pedagogy*'. This approach aimed to broaden the scope of the search while ensuring relevance to instructional contexts. Given the robustness and reputation of the Web of Science Core Collection database in housing high-quality scholarly material, the search strategy was tailored to this platform (Guo et al., 2024; Zou et al., 2021). Following the identification of pertinent search terms, the initial phase of the systematic review process commenced, resulting in the successful retrieval of 807 articles from the Web of Science. Table 1 provides a summary of these findings, illustrating the breadth of literature identified during this preliminary phase.

Screening

In the second phase of the study, the researchers rigorously screened the initial pool of 807 publications using stringent inclusion and exclusion criteria to minimize biases in the selection and review process. The criteria employed for selecting publications were meticulously defined as follows: (1) document type 'Article'; (2) publication date between 2022 and 2024; (3) publication language being English; (4) the journal are constrained to three categories of Educational Research based on Journal Citation Reports (JCRs): 'Education & Educational Research', 'Education, Special' and 'Education, Scientific Discipline'; (5) indexing in the Social Sciences Citation Index (SSCI); (6) publication by esteemed publishers such as Springer Nature, Elsevier, Taylor & Francis, Sage, Wiley and Emerald Group Publishing; (7) empirical nature of the study (qualitative, quantitative or mixed methods); (8) relevance to learning, instruction

Table 1. The search strings.

Database	Search strings
Wos	TS = 'generative AI*' or 'generative artificial intelligence*' or 'GenAI*' or 'Large language model*' AND 'education*' or 'teaching and learning*' or 'pedagogy*'

and pedagogy and (9) explicit descriptions provided regarding the research countries, sources of publication, research purposes, methodological approaches, educational levels of studies, application domains and GenAI platforms and apps.

Consequently, the following articles were excluded from the study: articles from review journals, books, conference proceedings and book chapters ($n=252$); articles published before 2022 ($n=1$); articles published in languages other than English ($n=7$); articles related to areas outside of educational studies ($n=337$); papers not indexed by the SSCI ($n=140$); articles published by publishers other than those specified ($n=10$); articles lacking empirical research ($n=25$); articles not directly related to teaching, learning or pedagogical practices ($n=4$); articles missing explicit descriptions of key research components ($n=2$) and duplicate studies ($n=1$). As a result, 779 articles were excluded based on these criteria, as detailed in [Table 2](#).

Eligibility and inclusion

During the third phase of the study, a total of 28 articles were meticulously gathered and subjected to rigorous assessment to determine their adherence to the predefined inclusion criteria. Key elements such as the title, abstract, research questions, problem statement, methodology and conclusions of each publication were meticulously scrutinized to ascertain their alignment with the study's objectives and suitability for inclusion in the review. To facilitate systematic categorization and analysis of pertinent information from the articles, a spreadsheet was prepared using Microsoft Word [(further details provided in the appendix ([Supplementary material](#)))]. This spreadsheet served as a structured framework for organizing essential details such as publication countries, sources, research purposes, methodologies employed, targeted educational levels, application domains explored and the specific platforms and applications utilized.

Subsequently, each of the 28 articles underwent thorough examination and analysis to extract relevant insights and information pertaining to the study's focus areas. [Figure 1](#) provides a visual representation of the systematic review analysis process, illustrating the meticulous scrutiny and categorization of data to facilitate comprehensive review and synthesis of findings. Through this meticulous approach, the study aimed to ensure the robustness and reliability of its analysis while fulfilling its overarching objectives.

Coding process

The systematic coding process employed in this review encompasses two pivotal components: the distinctive features of selected studies and the potential landscape of GenAI research in pedagogy. Drawing upon the techniques proposed by Li and Wong (2021), the coding framework for identifying distinctive features includes categories such as research locations, publication sources, research purposes, methodologies, educational levels, application domains, GenAI platforms and applications, among others. Additionally, the coding of the GenAI landscape was informed by the framework outlined by

Table 2. The inclusion and exclusion criteria.

Criterion	Inclusion	Exclusion
Document type	Article	Review journals, book, conference proceedings, book chapter
Timeline	Between 2022 and 2024	<2022
Language	English	Non-English
Research categories (JCRs)	Education & Educational Research; Education, Special; Education, Scientific Discipline	Other areas
Index	SSCI	Non-SSCI
Publisher	Springer Nature; Elsevier; Tator& Francis; Sage; Wiley; and Emerald Group	Other publishers
Type of research	Empirical research	Other type of research
Research direction	Teaching, learning and pedagogy	Other than these directions
Content of research	Descriptions of the research countries, sources publication, research purposes, methodological approaches and so on	Not specified of the research countries, sources publication, research purposes, methodological approaches and so on

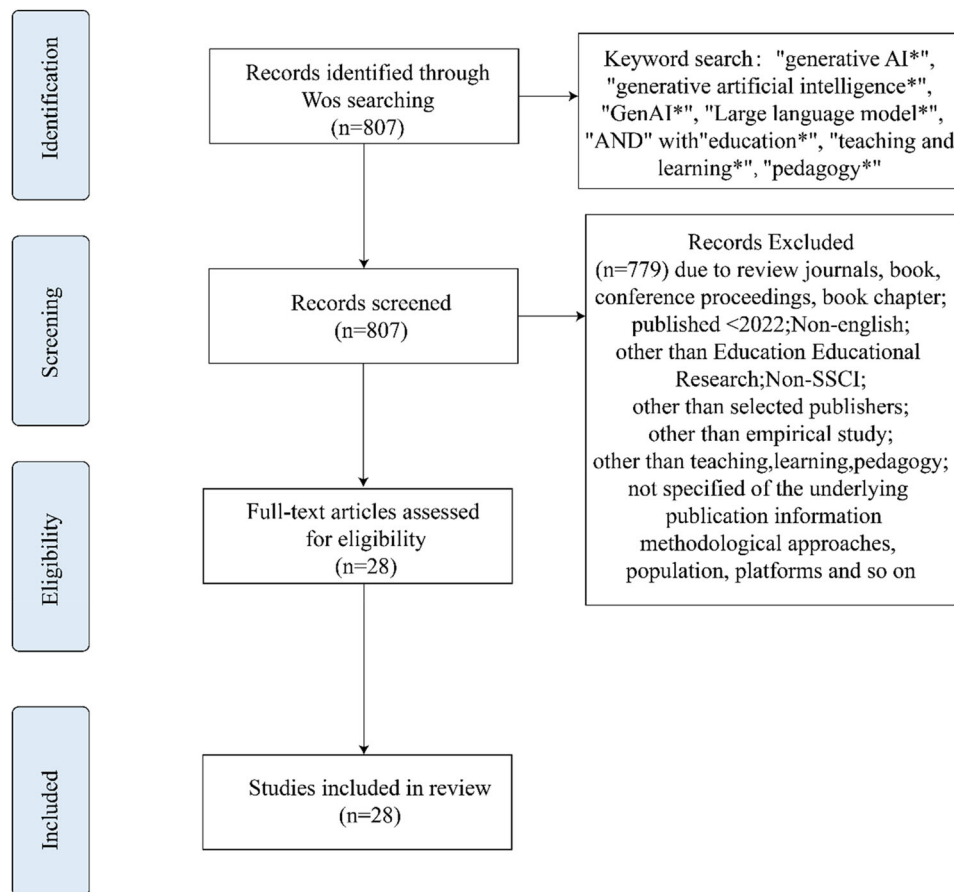


Figure 1. The sequential process of the review analysis.

Zainuddin et al. (2020). This framework prioritizes recurring themes to facilitate an analysis of future research trajectories in the field, offering insightful analysis and recommendations. Following individual coding of each of the 28 studies, the first and second researchers from the program team independently conducted coding based on the criteria and compared their findings. Substantial agreement was observed, with over 90% of their coding results aligning. In instances of disagreement, thorough discussions ensued until consensus was reached. Comprehensive inquiries and debates were undertaken to resolve any discrepancies, with involvement of a third author as needed to facilitate consensus-building. This meticulous approach ensured the robustness and reliability of the coding process, thereby enhancing the credibility and validity of the review's findings.

Data collection and analysis procedure

This study embarked on a structured approach aimed at assessing the potential landscape of GenAI in pedagogical research. Setting clear objectives at the outset, the study sought to comprehensively examine various facets including research countries, publication sources, research purposes, methodologies, educational levels, application fields, GenAI platforms and applications, prospective landscape and future trends. Utilizing the PRISMA screening methodology alongside a comprehensive keyword search ensured exhaustive data coverage while upholding the credibility of the study, thus validating the efficacy of this approach. Subsequently, an integrative review approach was employed to evaluate and explore the collected data. This approach encompasses the integration and evaluation of diverse research methods, including mixed methods, qualitative and quantitative approaches. As described by Whittemore and Knafl (2005), the integrative review approach involves the transformation of qualitative data into quantitative format and vice versa.

In this study, a qualitative investigation was conducted on the selected studies using thematic analysis to identify and categorize relevant themes and subthemes. Prior to analysis, data was meticulously collected, organized logically and key concepts were generated. Subsequently, a thorough analysis of the 28 papers was conducted to extract relevant content pertaining to the study topics. The information was then systematically arranged into major categories based on emergent concepts, ideas or themes. Ultimately, three primary themes were identified: the distinctive features of GenAI research, the prospective landscape of GenAI in pedagogy and trends for future research. Subthemes were further developed from these overarching themes, resulting in a total of 12 subthemes. This structured approach facilitated a comprehensive and nuanced exploration of the GenAI landscape in instructional research, providing valuable insights for further inquiry and development in this burgeoning field.

Findings

Distinctive features of the included studies (RQ1)

Research countries

In this section, we present the findings of this study, focusing on the distinctive features of the selected studies to provide a comprehensive overview of research on GenAI in learning and instruction. As illustrated in Figure 2, the articles analyzed in the review originated from institutions in 13 different countries or regions. Most of the studies were conducted in Hong Kong, China ($n=5$, 17.86%) and the United States ($n=5$, 17.86%), followed closely by Taiwan, China ($n=3$, 10.71%) and Vietnam ($n=3$, 10.71%). Mainland China ($n=2$, 7.14%), South Korea ($n=2$, 7.14%) and Saudi Arabia ($n=2$, 7.14%) were tied for third place. The remaining studies were carried out in Bangladesh ($n=1$, 3.57%), Germany ($n=1$, 3.57%), Malaysia ($n=1$, 3.57%), Poland ($n=1$, 3.57%), Sweden ($n=1$, 3.57%) and the United Arab Emirates ($n=1$, 3.57%). This distribution highlights the concentration of GenAI research in technologically advanced regions, where institutions are more likely to integrate AI innovations into instructional practices. Conversely, the limited presence of studies in other regions suggests an opportunity for extending research to diverse cultural settings to better understand GenAI's impact on teaching and learning.

Publication sources

Journals specializing in educational technology are indeed more inclined to publish research on GenAI in instructional contexts. As depicted in Figure 3, the *Education and Information Technologies*, published by Springer (IF, 4.8), emerges as the predominant source for GenAI in learning-focused research ($n=7$,

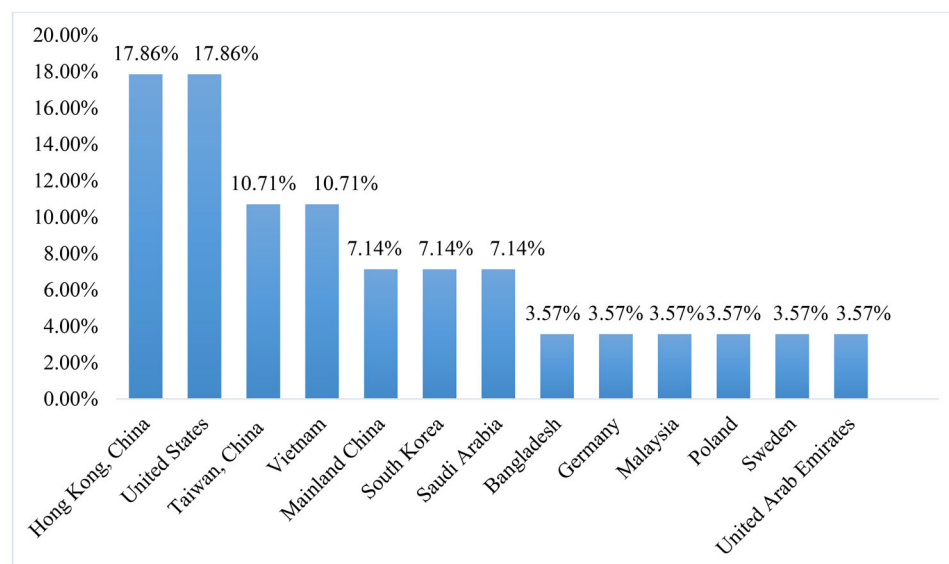


Figure 2. Article distribution by research countries.

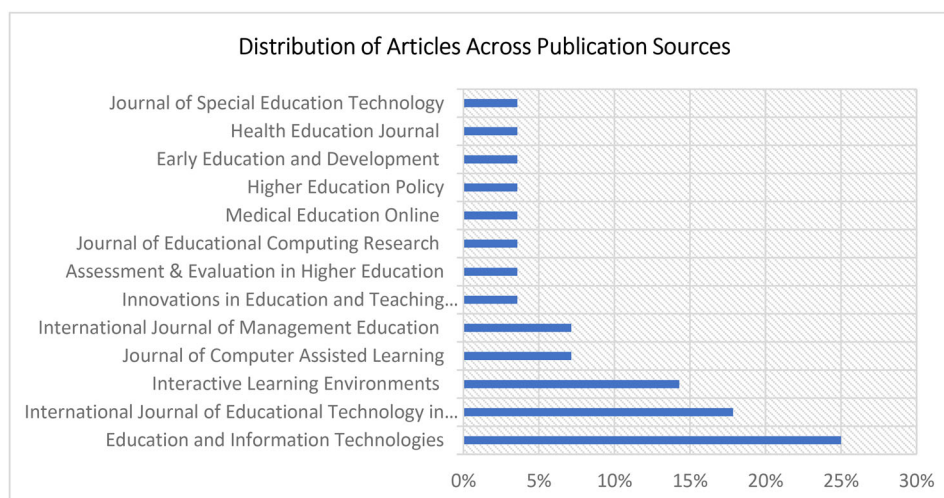


Figure 3. Distribution of articles across publication sources.

25%) of the publications. The *International Journal of Educational Technology in Higher Education*, also by Springer (IF, 8.6), follows closely with five studies ($n=5$, 17.86%). *Interactive Learning Environments* by Taylor & Francis Group (IF, 3.7) published four articles ($n=4$, 14.29%). This is succeeded by the *Journal of Computer Assisted Learning* from Wiley (IF, 2.36) and the *International Journal of Management Education* from Elsevier (2.1), each contributing two articles ($n=2$, 7.14%). The remaining studies were distributed across various journals: *Innovations in Education and Teaching International* (IF: 1.9) and *Assessment & Evaluation in Higher Education* (IF: 1.97), both by Taylor & Francis Group; *Journal of Educational Computing Research* by Sage (IF, 2.35); *Medical Education Online* by Taylor & Francis Group (IF: 1.83); *Higher Education Policy* by Springer (IF: 1.18); *Journal of Special Education Technology* (IF: 2.0) and *Health Education Journal* (IF: 0.49), both by Sage; and *Early Education and Development* by Taylor & Francis Group (0.93), each presenting one article ($n=1$, 3.57%). These findings suggest that the field has focused primarily on technology-enhanced pedagogical practices, including adaptive learning and AI-supported teaching innovations, rather than purely theoretical discussions or broad policy concerns.

Research purposes

The research purposes within the field of GenAI in learning and instruction exhibit considerable diversity. As depicted in Figure 4, the most prevalent aim observed in half of the studies ($n=14$, 50%) was to investigate stakeholders' perceptions and experiences. Following this, seven articles, constituting 25% of the studies ($n=7$, 25%), were dedicated to examining the effectiveness or impact of GenAI in instructional settings, underscoring it as another commonly pursued research objective. Additionally, two studies ($n=2$, 7.14%) were focused on developing a model related to GenAI, while an equal number ($n=2$,

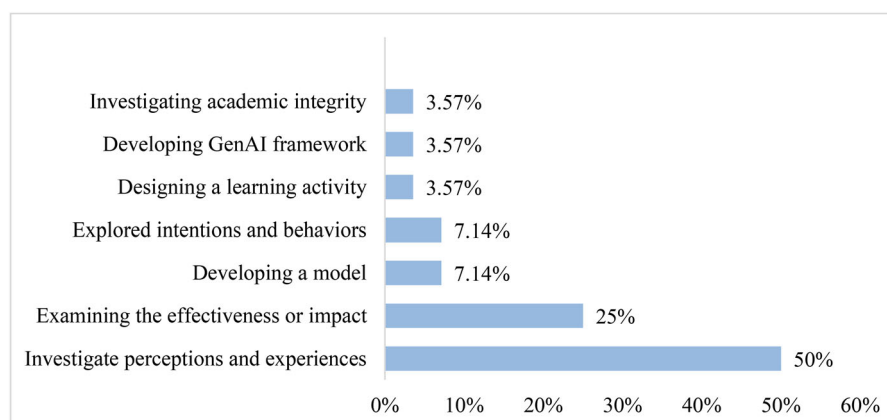


Figure 4. Article distribution by research purposes.

7.14%) delved into stakeholders' intentions and behaviors concerning GenAI. The remaining research endeavors were distributed among designing a learning activity, developing a GenAI framework and investigating academic integrity issues related to GenAI, with each of these objectives being the focal point of one study ($n = 1$, 3.57%). These trends emphasize a growing interest in understanding how GenAI reshapes traditional teaching methods and enhances learner engagement through adaptive technologies.

Methodological approaches

In terms of research methods, 11 studies (39.28%) employed a mixed-methods approach, 10 (35.71%) utilized a quantitative research approach, and 7 (25.00%) adopted a qualitative approach (Figure 5). Further examination of data collection methods revealed that questionnaire surveys and pre- and post-tests were predominantly used in gathering quantitative research data. Questionnaires were generally employed to capture stakeholders' perceptions of GenAI in instructional contexts. Meanwhile, pre- and post-test data were often utilized to evaluate the impact of GenAI usage on student academic performance. For qualitative data collection, interviews and open-ended question surveys emerged as the most popular techniques. Additional methods, including drawing tasks, chatbot usage logs, document analysis, focus group discussions and observations, were also employed for qualitative data acquisition. This combination underscores the importance of both quantitative and qualitative data in assessing the instructional value of GenAI tools. Comprehensive details can be reviewed in the appendix (Supplementary material).

Educational level of studies

As depicted in Figure 6, the most frequently analyzed educational level in the studies was higher education, accounting for 71.42% ($n = 20$). This was followed by primary education at 10.71% ($n = 3$). Secondary education and preschool each occupied the third position, both comprising 7.14% ($n = 2$), while kindergarten represented the least examined level, making up 3.57% ($n = 1$). These findings suggest that the flexibility and self-regulation inherent in university-level instruction make it a fertile ground for experimenting with GenAI. In contrast, the limited application in earlier educational stages points to potential areas for further research on adapting GenAI tools to diverse learner needs and curricula.

Application domains

As depicted in Figure 7, a comprehensive review of 28 research publications elucidated the diverse array of application domains. Language education and general higher education emerged as the predominant foci, each constituting 25% ($n = 7$). Subsequently, early childhood education (ECE), medical education and education policies garnered attention, each comprising 7.17% ($n = 2$). Additional areas of inquiry encompassed academic research, STEAM education, mathematics education, physics education, special education and entrepreneurship education, with each domain representing 3.57% ($n = 1$). Notably, a further 7.17% ($n = 2$) of the research delved into other domains, thus illustrating the breadth and depth of investigation within the field. Language education and general higher education emerged as central

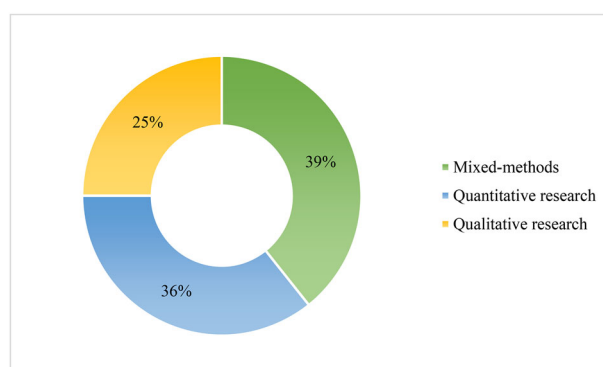


Figure 5. Article distribution by methodological approaches.

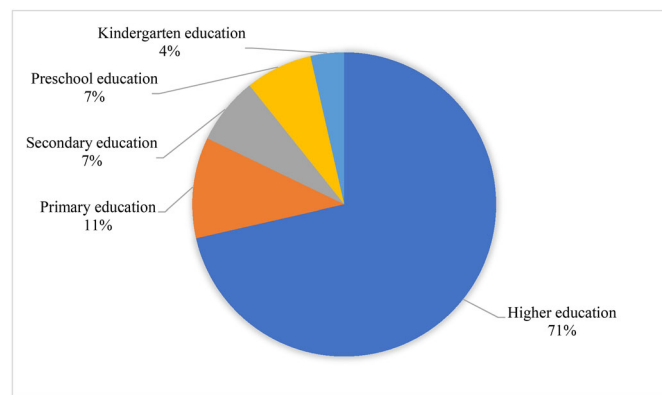


Figure 6. Article distribution by educational level of studies.

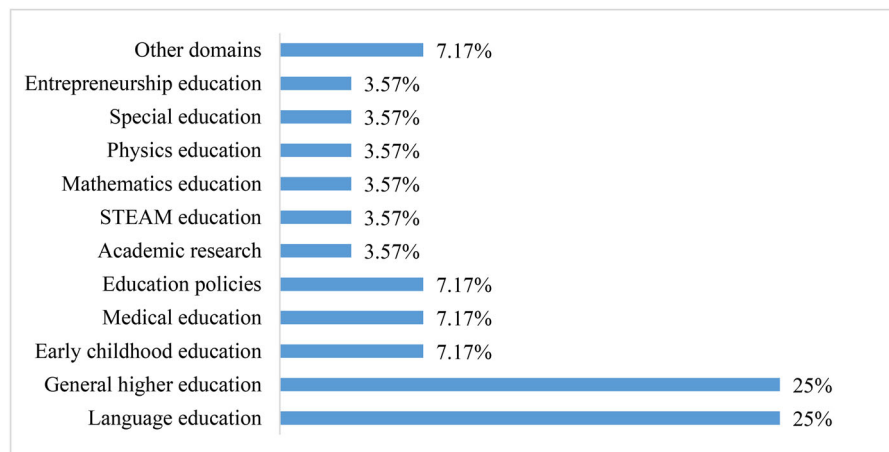


Figure 7. Article distribution by application domains.

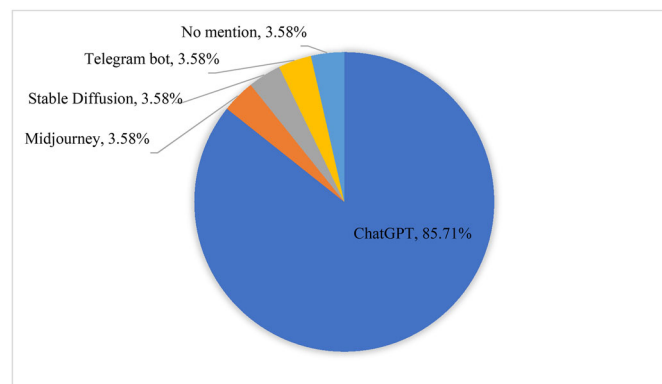


Figure 8. Article distribution by GenAI platforms and apps.

areas of study, illustrating GenAI's natural fit for text-based tasks. This prominence reflects ongoing interest in evaluating GenAI's immediate utility for writing, feedback and broader instructional processes.

GenAI platforms and apps

As illustrated in [Figure 8](#), our examination of 28 research papers revealed that a substantial majority (85.71%, $n = 24$) employed ChatGPT as the primary platform for teaching and learning. The remaining studies were distributed across Midjourney, Stable Diffusion and a Telegram bot, with each platform being the subject of one study (3.58%, $n = 1$). Moreover, one study (3.58%, $n = 1$) did not specify the platform utilized. This trend reflects ChatGPT's dominance in enabling personalized learning and interactive dialogue for pedagogical purposes. However, the limited exploration of alternative platforms

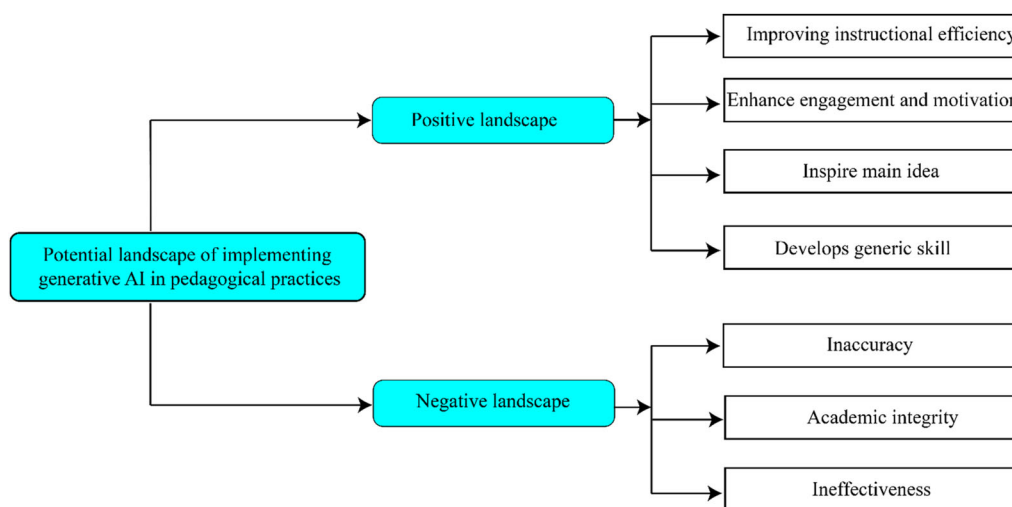


Figure 9. Thematic map.

suggests a need for broader investigations into the comparative advantages of different GenAI tools in teaching contexts.

Potential landscape of implementing GenAI in pedagogical practices (RQ2)

This section explores the research question regarding the potential applications of GenAI across various instructional domains. A comprehensive review of 28 empirical articles identified four key positive effects: augmenting instructional efficiency, fostering engagement and motivation, catalyzing core idea formation and developing generic skills. Additionally, three main challenges were noted, focusing on potential inaccuracies, academic integrity and limited efficacy. Figure 9 presents these findings in a thematic map, while the appendix (Supplementary material) offers more in-depth insights.

Improve instructional efficiency

Learning efficiency refers to enhancements in performance accuracy and speed relative to time spent learning, making it a critical measure of academic progress. It involves both improved performance and shorter learning durations. Essentially, higher efficiency means students can reach proficiency faster (Bruce, 2004). Effective instructional strategies enable students to grasp complex concepts swiftly while retaining knowledge for longer periods (Field, 2005). According to Field (2005), efficiency hinges on the strategic use of varied teaching methods and tools.

The integration of GenAI in instruction, leveraging advanced technologies like natural language processing and machine learning, offers substantial potential for improving learning efficiency. Such technology supports more effective teaching by providing immediate feedback, personalizing materials for individual learners, reducing burdens on both educators and students and increasing productivity (Ray, 2023).

Multiple studies illustrate how GenAI can significantly boost productivity and conserve time in instructional settings, thereby improving learning efficiency (Al-Zahrani, 2024; Mohamed, 2024; Vecchiarini & Somià, 2023). For example, Al-Zahrani (2024) found that these technologies are perceived as efficient tools for data analysis, collaboration and enhancing research outputs among higher education faculty. Users commended these tools' capacity to improve productivity and broaden the scope of research projects.

Similarly, Vecchiarini and Somià (2023) analyzed ChatGPT's role in higher education entrepreneurship classes. Their findings highlighted ChatGPT's pivotal function in streamlining routine tasks and cutting back instructor workload, illustrating how GenAI can strengthen instructional practices in entrepreneurial contexts. In another study, Rakap (2024) examined how ChatGPT assists novice special education teachers in creating high-quality Individualized Education Program (IEP) goals more quickly, suggesting that

Table 3. Summary of findings on improving instructional efficiency in GenAI research.

Instructional efficiency	Studies
GenAI tools significantly improve research efficiency, collaboration and productivity. Researchers report positive experiences, highlighting that these tools not only extend research capabilities but also enhance efficiency with few technical challenges.	Al-Zahrani (2024)
ChatGPT improves efficiency and productivity by automating repetitive tasks and reducing workload, suggesting its potential to streamline processes in entrepreneurship education.	Vecchiarini and Somià (2023)
ChatGPT significantly enhanced the quality of IEP goals for prospective special education teachers, who also spent less time on them compared to their counterparts who did not use the tool. Incorporating ChatGPT into educational practice offers a way to boost the efficiency of goal setting for educators.	Rakap (2024)

incorporating ChatGPT into teaching methods can enhance goal-setting efficiency. Table 3 summarizes the major findings on the role of GenAI in improving instructional efficiency.

Enhance engagement and motivation

Engagement and motivation are central to academic success, shaping class participation, instructional objectives, the enjoyment of learning and overall achievement (Collie & Martin, 2019). Teachers prioritize these elements due to their significant influence on student outcomes (Peng, 2021). Integrating technology into pedagogical practices has demonstrated benefits in boosting student engagement, peer collaboration and instructor-learner interactions (Schindler et al., 2017).

Studies like Muñoz et al. (2023) show how ChatGPT and other advanced technologies can make instruction more dynamic and participatory, thereby increasing student enthusiasm. Lai et al. (2023) propose that chatbots can heighten both cognitive and emotional motivation while facilitating seamless knowledge acquisition. Likewise, Ait Baha et al. (2023) found that chatbots allow students to learn at their own pace, reducing stress, saving time and bolstering motivation.

Our analysis emphasizes GenAI's significant effect on promoting intrinsic motivation and engagement. Students can explore unresolved problems with less anxiety, making learning more enjoyable and reducing feelings of inadequacy. Research by Wu et al. (2024), U. Lee et al. (2024) and Mohamed (2024) confirms that GenAI effectively boosts engagement and motivation, even in the absence of external pressures. For instance, Wu et al. (2024) utilized the Intelligent Learning Aid (CILA), a ChatGPT-based tool, to improve knowledge creation and self-regulation in mathematics classes. Implementing CILA significantly raised intrinsic motivation levels and enhanced cognitive, behavioral and emotional engagement. Similarly, U. Lee et al. (2024) demonstrated that incorporating image-GenAI in art education increased male students' engagement by reducing disparities in talent and interest. Additionally, Mohamed (2024) argues that ChatGPT enriches language learning through interactive dialogues, further heightening student motivation. Table 4 encapsulates these findings, highlighting GenAI's potential to enhance learner engagement and motivation.

Table 4. Summary of findings on enhancing engagement and motivation in GenAI research.

Enhance engagement and motivation	Studies
CILA significantly boosts learners' intrinsic motivation and decreases their amotivation by eliminating the fear of unresolved problems, enhancing enjoyment in learning and reducing feelings of helplessness. It also notably improves cognitive, behavioral and emotional engagement among learners.	Wu et al. (2024)
The AI-integrated art program improves student engagement, widens participation and narrows the skill and interest gap in art. Using GenAI in STEAM art classes positively changes students' views on AI-assisted picture creation and boosts motivation, especially in male students.	U. Lee et al. (2024)
The capacity of ChatGPT to produce responses that mimic human interaction and participate in dialogue with learners has the potential to boost motivation and engagement, thereby rendering the process of learning languages more enjoyable and rewarding	Mohamed (2024)

Inspire main idea

According to Wegerif’s (2010) dialogic theory, creativity emerges from a ‘dialogic space’, where varied viewpoints intersect, allowing novel ideas to flourish. This space expands with the inclusion of more voices, increasing the likelihood of innovative insights. When AI-generated responses serve as supplementary voices, learners can broaden this dialogic space, transcending existing concepts and inspiring unique perspectives (Scardamalia & Bereiter, 2016). As Filippi (2023) notes, digital tools significantly aid idea generation. Moreover, Girotra et al. (2023) suggest that GenAI often produces higher-quality ideas than humans.

Consistent with these assertions, recent research shows that students and teachers recognize GenAI as a valuable resource for idea generation and inspiration (Barrett & Pack, 2023; C. K. Chan & Hu, 2023; ElSayary, 2024). C. K. Chan and Hu (2023) highlight university students’ enthusiasm for GenAI platforms like ChatGPT, particularly for writing and brainstorming. Students can leverage AI to jump-start the idea-development process by posing targeted questions. Similarly, Barrett and Pack (2023) note that GenAI is most advantageous during the brainstorming and outlining stages of writing rather than as an automated completion tool. ElSayary (2024) emphasizes how teachers benefit from ChatGPT’s capacity to organize lesson plans and generate prompts, enriching instructional planning. Table 5 details key findings on GenAI’s role in inspiring main ideas within teaching contexts.

Develops generic skill

Generic skills such as critical thinking, collaboration, communication, argumentation and problem-solving are widely recognized as fundamental to twenty-first-century instruction and literacy (Tuononen et al., 2022). These competencies are core elements of higher education curricula, designed to enhance learning and prepare students for future careers (Tuononen et al., 2019). Scholars in GenAI argue that it strengthens students’ generic skills and shapes teachers’ perceptions of these competencies. For instance, Su and Yang (2023) investigated kindergarten teachers’ perspectives on ChatGPT, examining its benefits, challenges and required support. Teachers noted that ChatGPT simplifies lesson planning and enriches teaching strategies. Furthermore, such tools are believed to foster essential skills like creativity and problem-solving.

Luo et al. (2024) interviewed six experts from China and the United States to explore ChatGPT’s role in ECE. They found that ChatGPT enhances children’s conversational and social abilities, aiding in learning social cues, manners and empathy. The study underscores the importance of creativity, data literacy and critical thinking in modern classrooms. Chiu (2024) investigated the influence of GenAI on teaching, assessment and administration, involving 88 educators. The findings emphasize the increasing focus on generic skills, such as lifelong learning and twenty-first-century competencies, even in traditionally knowledge-centered cultures. Table 6 highlights relevant findings regarding generic skills in GenAI-focused instructional research.

Challenges of applying GenAI in education

While GenAI offers clear advantages for students and instructors, it also presents several notable challenges. C. K. Chan and Hu (2023) and Ding et al. (2023) emphasize GenAI responses’ fluency and

Table 5. Summary of findings on inspiring main ideas in GenAI research.

Inspire main idea	Studies
At times, students struggle to come up with ideas or inspiration. In these situations, by entering a question related to the writing topic, the output from AI can act as an initial catalyst for developing and broadening their ideas.	C. K. Chan and Hu (2023)
Both students and teachers view the use of GenAI as more favorable during the initial phases of the writing process, such as brainstorming and outlining, compared to its use in later stages. This indicates that GenAI is considered more appropriate for writing when it serves a supportive function in generating and organizing ideas, rather than being used as an automated tool for completing writing tasks.	Barrett and Pack (2023)
ChatGPT can help in organizing and structuring lesson plans, offering ideas, prompts and inspiration to create engaging and impactful lesson plans	ElSayary (2024)

Table 6. Summary of findings on develops generic skills in GenAI research.

Generic skill	Studies
ChatGPT can aid educators in crafting and refining lesson plans, offering resources and ideas to enhance their teaching methods. Utilizing AI tools such as ChatGPT can also promote the growth of critical thinking, creativity and problem-solving abilities in students, skills that are crucial for success in the twenty-first century.	Su and Yang (2023)
ChatGPT plays a role in improving conversational and social skills. By engaging in interactive dialogues, children can acquire social cues, etiquette and empathy, which are crucial for successful real-life communication. Additionally, certain abilities and knowledge areas, including critical thinking, data literacy, creativity, lifelong learning and interdisciplinary understanding, have gained increased importance.	Luo et al. (2024)
The introduction of GenAI has positively influenced teachers' perceptions of students' generic skill, underlining the increased significance of these skills in the era of GenAI.	Chiu (2024)

Table 7. Summary of findings on the challenges of implementing GenAI in pedagogical practices.

Findings	Studies
GenAI can swiftly generate responses that seem fluent and human-like; however, it cannot always assure their accuracy. As AI-driven conversations grow more common, the lack of transparency might hinder public trust.	C. K. Chan and Hu (2023), Ding et al. (2023)
Students and teachers generally expressed their concerns regarding its threats to academic integrity.	Farazouli et al. (2024), Yan (2023)
GenAI has not significantly enhanced learning outcomes and has sometimes negatively affected them. Consequently, its adoption in educational environments should be approached with caution, considering potential disadvantages.	Escalante et al. (2023), Niloy et al. (2024)

human-like qualities while raising concerns about reliability, transparency and accuracy. These observations indicate that GenAI should be used cautiously in instruction to mitigate misunderstandings. Policies must promote ethical and responsible AI usage to support learning effectively.

Concerns about academic integrity also arise. Yan (2023) examined how GenAI might circumvent plagiarism detection in writing classes and recommended careful application, particularly for second-language learners, to maintain ethical standards. Similarly, Farazouli et al. (2024) explored undergraduate instructors' perspectives on using GenAI in exam settings, observing occasional duplications while reinforcing its limitations and potential for misuse.

Moreover, questions remain regarding the overall effectiveness of GenAI. Escalante et al. (2023) compared AI feedback for English as a New Language (ENL) learners with traditional instruction and found it lacking. Niloy et al. (2024) revealed that GenAI negatively impacted university students' creative writing abilities, particularly their capacity to produce distinctive and accurate content. These findings highlight the need for cautious and balanced integration of GenAI, especially in highly creative or sensitive areas of instruction. A hybrid model that combines AI tools with human guidance may offer a practical approach. Table 7 summarizes the main concerns associated with GenAI in instructional contexts.

Discussion and recommendations for future work (RQ3)

This article evaluates empirical research on GenAI across various instructional contexts. It addresses academic gaps and provides informative and practical insights for future research, shedding light on how GenAI may reshape classroom practices in the digital age. In contrast to earlier systematic reviews that broadly examined AI in education (Ogunleye et al., 2024; Xiaoyu et al., 2024; Yusuf et al., 2024), this study isolates GenAI as a unique pedagogical tool, allowing for a more precise examination of its evolving applications. By synthesizing findings exclusively from post-2022 empirical studies, this review identifies emergent instructional trends, including how educators and learners navigate the ethical, creative

and cognitive challenges associated with GenAI tools. This level of specificity enhances the academic discourse by delineating GenAI's distinct role in contemporary educational environments.

China and the United States are leading the way in GenAI research on learning and instruction. Countries in Asia have produced significantly more research in this field than the combined total from the Americas, Europe, Africa and Oceania. To broaden research efforts on the instructional applications of GenAI, scholars in these regions must enhance their research and publication activities. Collaborating with colleagues from other countries is also important, as the current findings may not fully capture perspectives from non-SSCI-indexed research. Countries in Asia have produced significantly more research in this field than the combined output from the Americas, Europe, Africa and Oceania. To expand research efforts on the instructional applications of GenAI, scholars in these regions should enhance their research and publication activities. Collaboration with colleagues from other countries is crucial, as current findings may not adequately reflect perspectives from non-SSCI-indexed research. Following this recommendation could foster more equitable growth in the industry overall and offer valuable insights beyond those found in Asian countries.

Recent inquiries (Balzan et al., 2024; Kim & Koo, 2024; L. Zhang & Xu, 2025) have also reaffirmed the importance of evaluating GenAI's real-world impact on teaching and learning through diverse methodological lenses, highlighting the ongoing need for robust evidence-based practices. This broader approach not only strengthens the research landscape but also ensures that GenAI's benefits are comprehensively understood and effectively implemented across various educational contexts.

Publications focusing on educational technology are paying considerable attention to GenAI research in pedagogical settings. Key sources include the journals *Education and Information Technologies* and *International Journal of Educational Technology in Higher Education*. Scholars should prioritize reputable journals in their field, selecting those best suited to their target audience to effectively disseminate their findings. Through participation in transdisciplinary publications and non-traditional publishing platforms, researchers have the opportunity to significantly increase the dissemination of their work. Multidisciplinary partnerships have the potential to enhance the impact of research across various academic communities and facilitate broader dissemination in technology-enhanced learning and related disciplines.

When it comes to research on GenAI in pedagogical contexts, the majority of emphasis has been on stakeholder perspectives and the effectiveness of this technology. However, there is limited research on GenAI-focused experimental and comparative studies, teaching methods and instructional activities. The future direction of this field lies in GenAI serving as both a pedagogical and technological tool, combining artificial intelligence with human teaching to enhance classroom instruction. Future research should assess how effectively GenAI collaborates with educators to improve instructional outcomes, compare differences among various GenAI tools and develop novel teaching methods. It is imperative to optimize GenAI's potential and its impact on learning achievements.

Researchers have employed mixed methods, quantitative methods and qualitative methods to study the widespread use of GenAI in instructional settings. However, longitudinal studies, particularly those focusing on long-term impacts on academic performance, retention and learning outcomes, are rare. Escalante et al. (2023) is one of the few studies that has undertaken a long-term investigation, highlighting the need for more research in this area. Therefore, we recommend that future studies pay closer attention to longitudinal designs to better understand GenAI's long-term pedagogical potential, consistent with the findings of Bahroun et al. (2023), which emphasized the importance of longitudinal studies for GenAI research in instruction.

GenAI and its use in higher education have been the primary focus of instructional research. GenAI combined with self-regulated learning strategies is particularly suitable for university students due to their increasing self-governance and self-directed learning (Lai et al., 2023). This research demonstrates that GenAI can work effectively with self-regulated learning. In contrast, research on GenAI in vocational education is lacking. Given its relevance, GenAI has the potential to enhance vocational instruction. Therefore, future research should prioritize examining the effectiveness of the technology in vocational settings to fully realize GenAI's potential in various pedagogical contexts.

GenAI has been extensively studied in the domains of linguistics (Yan, 2023), general higher education (Adams et al., 2024) and subject-specific instruction (Ding et al., 2023), highlighting its remarkable

strengths in these areas. However, there is a notable gap in GenAI research related to Sustainable Development Goal 4, which emphasizes ‘quality education’ (Al-Sharafi et al., 2023). Given this gap, we recommend that future studies delve into GenAI’s impact on and contributions to ‘quality education’ and specifically to achieving Sustainable Development Goal 4.

Regarding GenAI platforms and applications, ChatGPT stands out as a leader in the GenAI domain, significantly impacting learning by personalizing instruction and automating content creation. Its role highlights the potential to meet diverse pedagogical needs and foster inclusivity (Kasneci et al., 2023). However, an exclusive focus on ChatGPT may overlook the broader educational technology landscape. The complexity of the field, along with the varied needs of learners, necessitates the exploration of multiple GenAI tools. Aydın and Karaarslan (2023) emphasize the need to examine a range of GenAI tools beyond ChatGPT, as different platforms offer unique strengths in academic research and writing, reinforcing the importance of a diverse AI ecosystem in education.

Therefore, adopting a comprehensive research strategy that evaluates the strengths and applications of different tools is crucial. Expanding research beyond a single platform is key to fully understanding GenAI’s role in enhancing instruction. We recommend that future research critically assess GenAI’s multidisciplinary and interdisciplinary applications alongside comparative studies that evaluate its collaboration with and comparison to other instructional technologies across various platforms, applications and international contexts. Additionally, exploring GenAI’s role in enhancing ‘quality education’ and contributing to Sustainable Development Goal 4 is essential.

In addition to these gaps, the broader ethical and social dimensions of GenAI implementation also warrant more in-depth examination. Firstly, the digital divide is a pressing concern: even though GenAI tools promise efficiency, personalized feedback and real-time support, schools and communities with limited technological resources or weak internet infrastructure may find them inaccessible. This uneven access risks widening current educational inequalities (Bulathwela et al., 2024). Therefore, policymakers and educational institutions must consider investments and interventions to ensure equitable access to hardware, software and teacher training in lower-resourced settings.

Secondly, as GenAI systems become more prevalent, overreliance on automated feedback can pose challenges for learners’ critical thinking and autonomy. While AI-driven assistance often boosts motivation and streamlines tasks, it may promote learners’ dependence on technology and potentially trigger ‘metacognitive laziness’ (Fan et al., 2025). Future research should explore integrated instructional frameworks or rubrics that blend GenAI’s rapid feedback with reflective exercises to preserve students’ active engagement and ownership of their learning processes.

Lastly, ethical issues, including data privacy, algorithmic bias and the reliability of AI-generated content—demand careful attention. Although GenAI can produce remarkably fluent responses, these outputs still depend on the quality and scope of its underlying training dataset (Banh & Strobel, 2023). Biases embedded in these datasets may perpetuate stereotypes or produce culturally insensitive materials. In addition, privacy concerns arise when GenAI platforms collect and analyze user interactions at scale, underscoring the need for robust data-protection policies. Moving forward, studies should address how to develop, implement and monitor clear ethical guidelines for GenAI usage, ensuring that AI-driven instruction aligns with both educational objectives and societal values.

Limitations

This review is subject to several critical limitations. Firstly, it is based on a relatively small selection of empirical studies, comprising only 28 sources. Future research should incorporate a broader range of literature, including book chapters, conference papers and review journals, to achieve a more comprehensive perspective. Secondly, the data collection process was confined to a single database (Web of Science), which may have limited the breadth of the investigation. Therefore, future studies could benefit from consulting additional databases, such as Scopus or Google Scholar, to capture a more diverse range of publications.

Thirdly, the temporal scope of the analysis was not fully addressed, as most GenAI research has emerged toward the end of 2022 and has continued through 2023 and 2024. Future research in this area must assess how publication patterns evolve as research accelerates. Lastly, the focus of this review

was limited to the application of GenAI within pedagogical contexts. Investigating other domains could provide a more holistic insight into GenAI's broader potential impact. Lastly, the sources are confined to three categories of Educational Research based on the JCRs: '*Education & Educational Research*', which includes 27 articles, and '*Education, Special*', which contains 1 article. However, there are no related articles to review in the category of '*Education, Scientific Disciplines*'. Further reviews are expected to explore these three educational categories based on the JCRs.

Conclusion

This review of empirical research on GenAI in pedagogical practices offers a comprehensive examination of various facets, including research sites, publication sources, research aims, methodological designs, levels of instruction, application domains and the evolving landscape of GenAI. The findings underscore GenAI as an emerging trend poised to enhance instructional efficiency, amplify student engagement and motivation, spark idea generation and foster the development of generic skills.

While innovative educational technologies such as flipped classrooms and gamified learning have influenced teaching methods in the 21st century, they have not fundamentally transformed the predominantly teacher-led and classroom-focused instructional model. In contrast, GenAI can potentially disrupt this model by enabling the rapid production of high-quality learning content. This capability empowers students to learn autonomously, confront challenging questions without hesitation and engage in learning anytime and anywhere, heralding a significant shift toward more flexible and accessible learning experiences.

The adoption of GenAI is anticipated to enhance student engagement and elevate the quality of instruction, although challenges related to accuracy, academic integrity and efficacy persist. Consequently, GenAI should be employed as a supplementary instructional resource under educator guidance to maximize its benefits while minimizing shortcomings.

In this dynamic environment, educators play a pivotal role in guiding and shaping the transformative impact of GenAI on teaching and learning. However, questions remain regarding effective integration strategies for classroom practices. Maximizing GenAI's instructional value requires innovative teaching approaches that enrich the learning process and make it more enjoyable. Therefore, it is vital to equip educators with the knowledge and skills necessary for successful GenAI implementation through targeted professional development. In practice, stakeholders should establish guidelines for responsible AI use in education to ensure transparency, address bias and uphold academic integrity. Educators require dedicated and ongoing training to master AI-driven pedagogical tools, adapt them to various education levels and effectively personalize instruction. Equally important, policymakers must develop ethical standards and regulatory frameworks that protect student privacy while supporting continuous innovation. By aligning these strategies with institutional priorities and allocating necessary resources, schools can create a supportive ecosystem that fosters meaningful AI-enhanced learning experiences. Furthermore, additional research into practical frameworks and methods for GenAI integration can help ensure that its use positively influences all stakeholders' learning outcomes.

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